

$$\begin{aligned}
& \wedge \text{router\_ct} = 0 \\
& \wedge \text{shard\_ct} = 0 \\
& \wedge \text{tx\_status} = \\
& \quad [t \in \text{AllTxns} \mapsto \text{IF } t = \text{InitTx} \text{ THEN "Committed" ELSE "Inactive"}] \\
& \wedge \text{tx\_snapshot} = [t \in \text{AllTxns} \mapsto 0] \\
& \wedge \text{tx\_checkset} = [t \in \text{AllTxns} \mapsto \{\}] \\
& \wedge \text{tx\_buffer} = [t \in \text{AllTxns} \mapsto [k \in \text{Keys} \mapsto \langle \rangle]] \\
& \wedge \text{tx\_commit\_time} = [t \in \text{AllTxns} \mapsto 0] \\
& \wedge \text{store} = \\
& \quad [k \in \text{Keys} \mapsto \{[\text{val} \mapsto \text{InitVal}, \text{cts} \mapsto 0, \text{writer} \mapsto \text{InitTx}]]\}] \\
& \wedge \text{pc} = [t \in \text{TxnIds} \mapsto 1] \\
& \wedge \text{op\_seq} = [t \in \text{TxnIds} \mapsto \langle \rangle]
\end{aligned}$$

$$\begin{aligned}
\text{StartTx}(t) & \triangleq \\
& \wedge \text{tx\_status}[t] = \text{"Inactive"} \\
& \wedge \text{LET } \text{sess} \triangleq \text{Workload}[t].\text{session} \\
& \quad \text{lvl} \triangleq \text{Workload}[t].\text{level} \\
& \text{IN IF } \text{lvl} \in \{\text{"PC"}, \text{"SI"}, \text{"SER"}\} \\
& \quad \text{THEN } \wedge \text{tx\_snapshot}' = [\text{tx\_snapshot} \text{ EXCEPT } ![t] = \text{shard\_ct}] \\
& \quad \quad \wedge \text{UNCHANGED router\_ct} \\
& \quad \text{ELSE IF } \text{lvl} \in \{\text{"CC"}, \text{"PSI"}\} \\
& \quad \quad \text{THEN } \wedge \text{tx\_snapshot}' = [\text{tx\_snapshot} \text{ EXCEPT } ![t] = \text{client\_dep}[\text{sess}]] \\
& \quad \quad \quad \wedge \text{UNCHANGED router\_ct} \\
& \quad \text{ELSE } \text{RA} \\
& \quad \quad \wedge \text{tx\_snapshot}' = \\
& \quad \quad \quad [\text{tx\_snapshot} \text{ EXCEPT } ![t] = \text{client\_last\_commit}[\text{sess}]] \\
& \quad \quad \quad \wedge \text{UNCHANGED router\_ct} \\
& \wedge \text{tx\_status}' = [\text{tx\_status} \text{ EXCEPT } ![t] = \text{"Active"}] \\
& \wedge \text{UNCHANGED } \langle \text{client\_dep}, \\
& \quad \text{client\_last\_commit}, \\
& \quad \text{shard\_ct}, \\
& \quad \text{tx\_checkset}, \\
& \quad \text{tx\_buffer}, \\
& \quad \text{tx\_commit\_time}, \\
& \quad \text{pc}, \\
& \quad \text{store}, \\
& \quad \text{op\_seq} \\
& \rangle
\end{aligned}$$

$$\begin{aligned}
\text{MaxVersion}(\text{versions}, \text{max\_cts}) & \triangleq \\
& \text{LET } \text{valid} \triangleq \{v \in \text{versions} : v.\text{cts} \leq \text{max\_cts}\} \\
& \text{IN CHOOSE } v \in \text{valid} : \forall v2 \in \text{valid} : v.\text{cts} \geq v2.\text{cts}
\end{aligned}$$

$$\text{ExecuteOp}(t) \triangleq$$